Low-Level Design – Application Settings

TaskFlow – Intelligent Task Allocation & Tracking System

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Based on: HLD-Application-Settings v1.0, FRD-Application-Settings v1.0
Module: Settings · **Sub-module:** 06.2 Application Settings · **Code:** APP
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Document Control

Version	Date	Author	Summary of changes
1.0	31-Oct-2026	Jojo George	Initial version

Related documents: BRD · FRD · HLD · TDD · Specs

Document	Location
BRD – Application Settings	`../../../BRD/06-Settings/06.2-Application-Settings/BRD-Application-Settings.md`
FRD – Application Settings (FR-APP-01…23, UC-APP-01…03)	`../../../FRD/06-Settings/06.2-Application-Settings/FRD-Application-Settings.md`
HLD – Application Settings	`../../../HLD/06-Settings/06.2-Application-Settings/HLD-Application-Settings.md`
Overall LLD (stack, cross-cutting design, Auth)	`../../00-Overall/LLD-TaskFlow-Overall.md`
LLD – Reminder Settings (shares the `ApplicationSettings` table and repository)	`../06.1-Reminder-Settings/LLD-Reminder-Settings.md`
LLD – Notification (`NotificationPublisher` , in-app switch consumer)	`../../04-Communication/04.2-Notification/LLD-Notification.md`
TDD v2.0	`TDD-TaskFlow.md` §7.2.6 (`ApplicationSetting`), §7.5 (date, time, timezone)

1. Introduction

1.1 Purpose

Detailed design of the Application Settings sub-module (original BRD Screen 20 "System Settings"): the two controller actions, `ApplicationSettingsService`, the application clock `AppClock`, the validator,

the shared repository, the four setting rows it owns, the Angular screen and shell brand name, tests and FR-APP-nn traceability. Everything is taken from the code as built.

1.2 Scope & Conventions

- Covered: `ApplicationSettingsController` (2 actions), `ApplicationSettingsService`, `AppClock` / `IAppClock`, `UpdateApplicationSettingsRequestValidator`, `ApplicationSettingsRepository` (shared with REM), the keys `Application.Name`, `Application.Timezone`, `Application.DefaultPriority`, `Notification.InAppEnabled`, and the Angular `ApplicationSettingsComponent` plus `ApplicationSettingsService` (Angular, `appName` signal).
- Referenced only: consumers of `IAppClock` (Reminder, Dashboard, Workload, Report, Task, Mail, Assistant services) and `NotificationPublisher` (Notification LLD).
- Naming: `*Controller`, `I*Service` / `*Service`, `I*Repository` / `*Repository`, DTOs `*Dto` / `*Request`.
- API base path `/api/v1`; JSON camelCase; enums as strings (`JsonStringEnumConverter`); errors as ProblemDetails.

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2. Code Structure

2.1 Backend Folder Layout

```
TaskFlow.Api/
  Controllers/Settings/ApplicationSettingsController.cs
  Services/Settings/ApplicationSettings/General/IApplicationSettingsService.cs,
ApplicationSettingsService.cs
  Services/Settings/ApplicationSettings/Clock/IAppClock.cs, AppClock.cs
  Repositories/Settings/ApplicationSettings/IApplicationSettingsRepository.cs,
ApplicationSettingsRepository.cs  (shared with REM)
  Dtos/Settings/ApplicationSettings/ApplicationSettingsDtos.cs
  Dtos/Settings/ApplicationSettings/Validators/UpdateApplicationSettingsRequestValidator.cs
  Models/Entities/ApplicationSetting.cs, Models/Enums/TaskPriority.cs
  Common/Constants/SettingKeys.cs, Common/Utils/SettingValues.cs
  Config/ApplicationOptions.cs
  Persistence/Seeders/SettingsInitializer.cs
```

2.2 Frontend Folder Layout

```
src/app/modules/settings/
  settings.routes.ts                                        (route settings/application)
  models/settings.model.ts                                  (ApplicationSettings,
UpdateApplicationSettings)
  application-settings/pages/application-settings.component.{ts,html,css}
  application-settings/pages/application-settings.model.ts    (SettingsField)
  application-settings/services/application-settings.service.ts
src/app/layouts/shell.component.ts, layouts/nav-items.ts, shared/auth/auth.guard.ts
```

2.3 Layer Rules

Rule	Enforced by
Controller → Service → Repository only, through interfaces	`ArchitectureTests`

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Rule	Enforced by
Only repositories (and startup/seeders) touch `AppDbContext`; services commit through `IUnitOfWork`	`ArchitectureTests`
Controllers do not validate; FluentValidation runs in `ValidationFilter`	`ArchitectureTests`, global filter
API service in the module whose endpoints it calls (`modules/settings/...`), page in `pages/`	`npm run check:structure`

3. Cross-Cutting Design

Shared mechanisms are in the overall LLD (`../../00-Overall/LLD-TaskFlow-Overall.md`). Applied here:

Concern	Design
Validation	`UpdateApplicationSettingsRequestValidator` in the global `ValidationFilter`; 400 `ValidationProblemDetails` with `errors`
Error handling	`GlobalExceptionHandler` (3.1)
Transactions	One `IUnitOfWork.SaveChangesAsync` writes all four rows
Authorization	GET: `[Authorize]` (any signed-in user). PUT: `[RequirePermission(PermissionKeys.SettingsApplication)]` → `perm:settings.application.manage`
Logging	`AppClock` logs an error "Unknown application timezone '...'; using UTC." when the stored id cannot be resolved
Time & time zone	Timestamps UTC (`TimeProvider`); `IAppClock` answers "which calendar day is it" in the application timezone for all deadline comparisons (BR-17 in code comments)

3.1 Exception → HTTP Status

Exception	Status	When
Validation failure (`ValidationFilter`)	400	Name blank or too long, timezone blank or unknown, priority or switch missing/invalid
Policy failure	403	PUT without `settings.application.manage`
– (JWT bearer) / `Unauthorized()` in the action	401	Not signed in; PUT without a numeric user id claim
`ConcurrencyConflictException` / `UniqueConstraintException`	409	Save race / first-insert race on `IX_ApplicationSettings_Key`
`ArgumentException` ("Every application setting must be provided.")	500	Only if the validator is bypassed (not an `AppException`)
`TimeZoneNotFoundException` / `InvalidTimeZoneException` in `AppClock`	–	Caught: logged, `TimeZoneInfo.Utc` used (FR-APP-09)

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4. Module Designs

4.1 Module: Settings

4.1.1 Child module: Application Settings (APP)

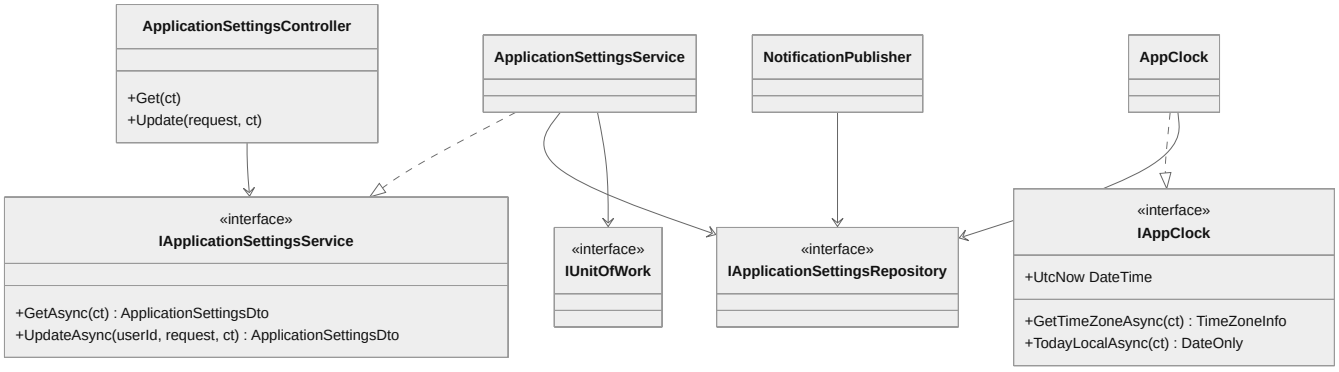
Class / file inventory

Layer	Type	File	Responsibility
Controller	`ApplicationSettingsController`	`Controllers/Settings/ApplicationSettingsController.cs`	`GET` / `PUT` `/api/v1/settings/application`; resolves `userId`
Service	`IApplicationSettingsService` / `ApplicationSettingsService`	`Services/Settings/ApplicationSettings/General/`	Typed view over four rows; defaults; upsert
Service	`IAppClock` / `AppClock`	`Services/Settings/ApplicationSettings/Clock/`	`UtcNow`, `GetTimeZoneAsync()`, `TodayLocalAsync()`
Repository	`IApplicationSettingsRepository` / `ApplicationSettingsRepository`	`Repositories/Settings/ApplicationSettings/`	Key/value reads and tracked rows (shared with REM)
Base	`BaseRepository<ApplicationSetting>`, `IUnitOfWork` / `UnitOfWork`	`Repositories/Base/`	`Add`, save, conflict mapping
DTOs	`ApplicationSettingsDto`, `UpdateApplicationSettingsRequest`	`Dtos/Settings/ApplicationSettings/ApplicationSettingsDtos.cs`	Contracts
Validator	`UpdateApplicationSettingsRequestValidator`	`Dtos/Settings/ApplicationSettings/Validators/`	Name, timezone, priority, switch rules
Entity	`ApplicationSetting`	`Models/Entities/ApplicationSetting.cs`	Key/value row
Enum	`TaskPriority`	`Models/Enums/TaskPriority.cs`	`Low`, `Medium`, `High`, `Critical`
Constants	`SettingKeys`, `SettingValues`	`Common/Constants/`, `Common/Utils/`	Keys, categories, defaults, formatting
Options	`ApplicationOptions`	`Config/ApplicationOptions.cs`	`Application:Timezone` (code default `UTC`)
Seeder	`SettingsInitializer`	`Persistence/Seeders/SettingsInitializer.cs`	Missing rows at start-up

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Layer	Type	File	Responsibility
Frontend page	`ApplicationSettingsComponent`	`modules/settings/application-settings/pages/application-settings.component.ts`	Screen, form, `requiredText` validator, `timezoneSuggestions`
Frontend service	`ApplicationSettingsService` (Angular)	`modules/settings/application-settings/services/application-settings.service.ts`	`get()`, `save()`, cached `appName` signal
Frontend shell	`ShellComponent`	`layouts/shell.component.ts`	Reads `appName`; calls `get()` at start
Frontend models	`ApplicationSettings`, `UpdateApplicationSettings`, `SettingsField`	`modules/settings/models/settings.model.ts`, `pages/application-settings.model.ts`	Types
Route / guard / menu	`SETTINGS_ROUTES`, `permissionGuard`, `buildNavigation()`	`settings.routes.ts`, `shared/auth/auth.guard.ts`, `layouts/nav-items.ts`	Access to `/settings/application`

Class diagram



API contracts

Controller `ApplicationSettingsController`, `[Route("api/v1/settings/application")]`, no class-level policy. Endpoints of this sub-module (facts.md §3): **2**.

#	Method	Route	Permission / policy	Request	Response	Status codes
1	GET	`/api/v1/settings/application`	`[Authorize]` (signed in)	–	`ApplicationSettingsDto`	200, 401
2	PUT	`/api/v1/settings/application`	`settings.application.manage`	`UpdateApplicationSettingsRequest`	`ApplicationSettingsDto`	200, 400, 401, 403, 409

DTOs

DTO	Field	Type	Notes
`ApplicationSettingsDto`	`Name`	`string`	`Application.Name`

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DTO	Field	Type	Notes
	`Timezone`	`string`	`Application.Timezone`
	`DefaultPriority`	`TaskPriority`	`Application.DefaultPriority`
	`InAppNotificationsEnabled`	`bool`	`Notification.InAppEnabled`
`UpdateApplicationSettingsRequest`	`Name`, `Timezone`	`string?`	Nullable so an omitted field fails validation
	`DefaultPriority`	`TaskPriority?`	
	`InAppNotificationsEnabled`	`bool?`	

The DTO carries nothing secret, which is why GET is open to every signed-in user (FR-APP-18, FR-APP-19).

Request / response example

```
// PUT /api/v1/settings/application
{ "name": "TaskFlow", "timezone": "Asia/Kolkata", "defaultPriority": "High",
"inAppNotificationsEnabled": true }
```

```
// 200 OK
{ "name": "TaskFlow", "timezone": "Asia/Kolkata", "defaultPriority": "High",
"inAppNotificationsEnabled": true }
```

```
// 400 Bad Request (unknown timezone)
{ "title": "One or more validation errors occurred.", "status": 400, "errors": { "Timezone":
["Unknown timezone."] } }
```

Service logic – `ApplicationSettingsService.GetAsync`

1. `GetValuesAsync(ApplicationKeys)` for `Application.Name`, `Application.Timezone`, `Application.DefaultPriority`, `Notification.InAppEnabled`.
2. `ReadString` (blank → fallback): name → `DefaultApplicationName` ("TaskFlow"); timezone → `ApplicationOptions.Timezone`, or `FallbackTimezone` ("UTC") when that is blank.
3. `ReadPriority`: `Enum.TryParse` (ignore case) + `Enum.IsDefined`, else `DefaultDefaultPriority` ("Medium").
4. `ReadBool`: `bool.TryParse`, else `DefaultInAppEnabled` (true). Returns `ApplicationSettingsDto` (FR-APP-17).

Service logic – `ApplicationSettingsService.UpdateAsync(userId, request)`

1. Validator already ran (400 on failure). Null field → `ArgumentException("Every application setting must be provided.")` (FR-APP-16).
2. `GetForUpdateAsync(ApplicationKeys)`; `now = TimeProvider.GetUtcNow().UtcDateTime`.

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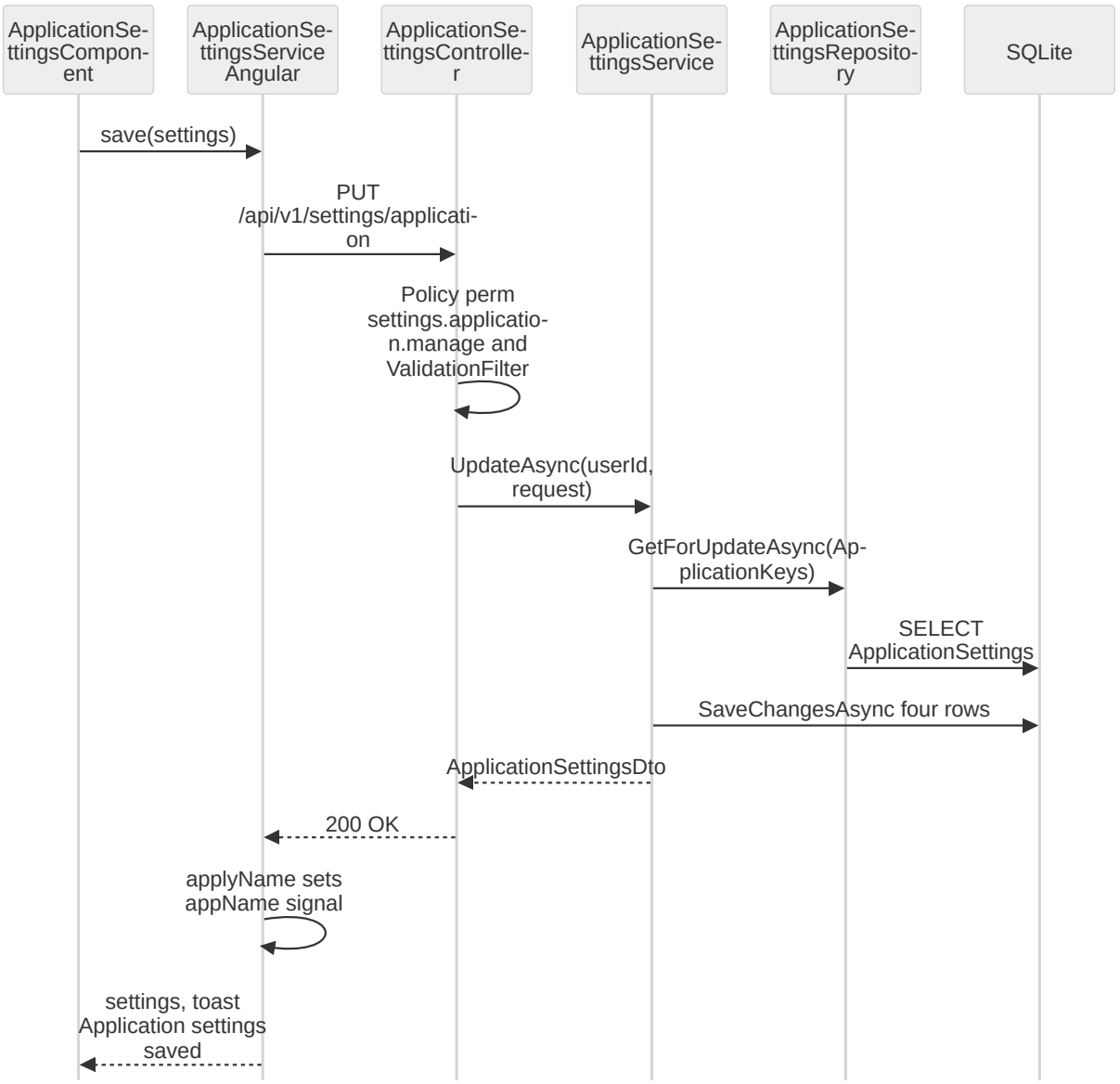
3. `Set(key, value, category)` creates a missing row and sets `Value`, `UpdatedByUserId`, `UpdatedAt` (FR-APP-20).
4. Values: `name.Trim()`; `timezone` as received; `defaultPriority.ToString()`; `SettingValues.Format(inAppEnabled)`. Categories: `Application` for the first three, `Notification` for the switch.
5. `IUnitOfWork.SaveChangesAsync`, then `GetAsync()`.

Service logic – `AppClock`

1. `UtcNow` = `TimeProvider.GetUtcNow().UtcDateTime`.
2. `GetTimeZoneAsync`: `GetValueAsync(Application.Timezone)`; blank → `ApplicationOptions.Timezone`; `TimeZoneInfo.FindSystemTimeZoneById(id)`; on `TimeZoneNotFoundException` / `InvalidTimeZoneException` → log error and return `TimeZoneInfo.Utc` (FR-APP-09).
3. `TodayLocalAsync`: `DateOnly.FromDateTime(TimeZoneInfo.ConvertTimeFromUtc(UtcNow, zone))`.
4. Read on every call (no cache), so a saved timezone applies to the next calculation without restart (FR-APP-08). Consumers: `ReminderService`, `DashboardService`, the analytics services and scopes, `WorkloadService`, `ReportService`, `TaskService`, `MailService`, `AssistantService`, `AssistantReadTools`.

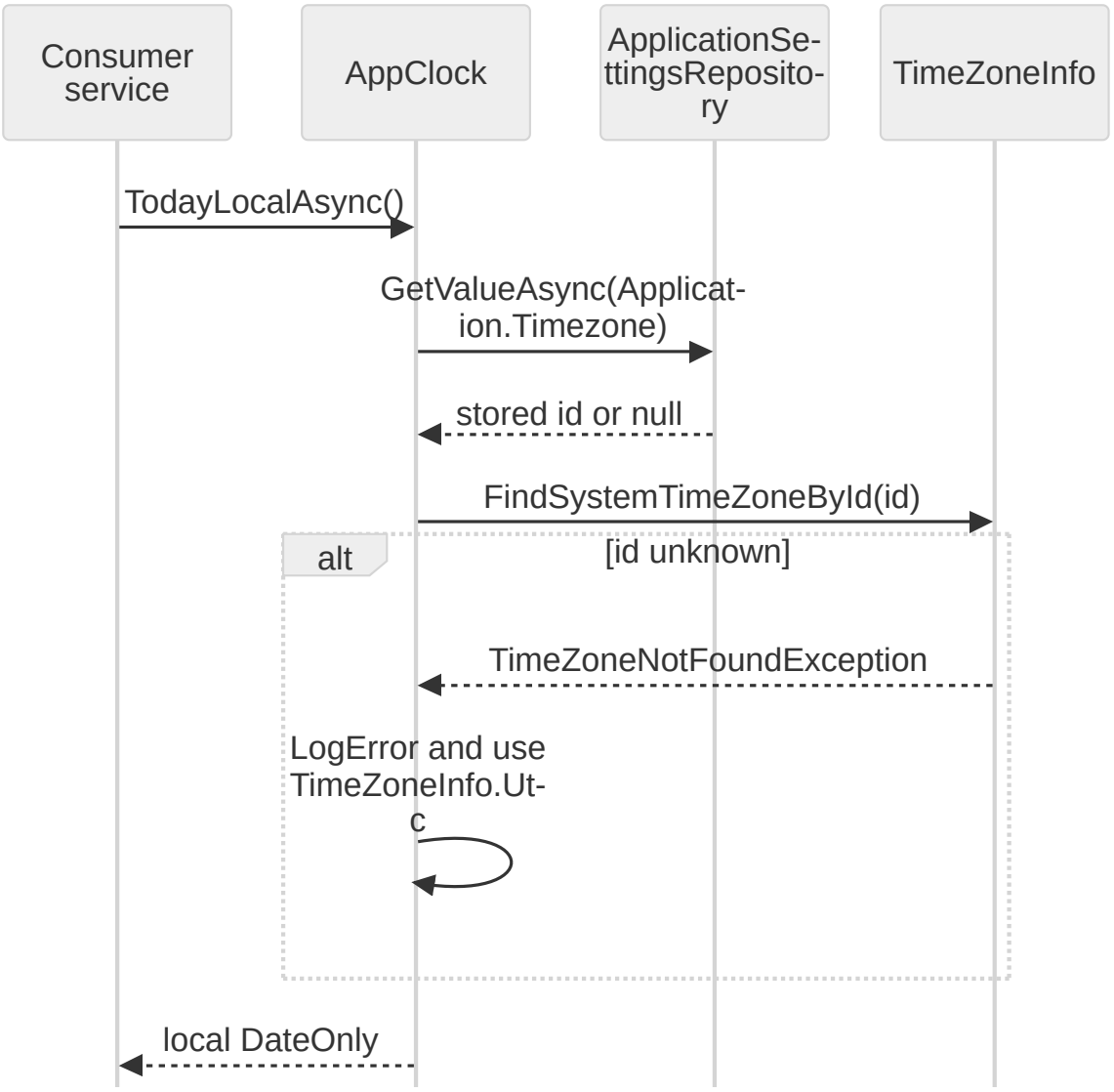
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Sequence diagram – save



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Sequence diagram – timezone resolution



Repository methods (`ApplicationSettingsRepository`)

Method	Query / purpose	Notes
`GetValuesAsync(keys)`	`AsNoTracking`, `Where(keys.Contains(Key))`, `ToDictionaryAsync(Key, Value)`	Used by `GetAsync`
`GetValueAsync(key)`	`AsNoTracking`, `Select(Value)`, `FirstOrDefaultAsync`	Used by `AppClock`, `NotificationPublisher`, `TaskCommentService`
`GetForUpdateAsync(keys)`	Tracked, `ToDictionaryAsync(Key)`	Used by `UpdateAsync`
`Add(entity)` (base)	Stage a new row	`BaseRepository`

Validation rules (`UpdateApplicationSettingsRequestValidator`)

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Field	Rule	Message
`Name`	`NotEmpty()`	"Application name cannot be blank."
`Name`	`MaximumLength(100)` (checked before trimming)	FluentValidation default
`Timezone`	`NotEmpty()`	"Timezone cannot be blank."
`Timezone`	`Must(BeKnownTimezone)` (`TimeZoneInfo.FindSystemTimeZoneById`; Windows or IANA ids, as the host OS recognises)	"Unknown timezone."
`DefaultPriority`	`NotNull()`, `IsInEnum()`	FluentValidation default
`InAppNotificationsEnabled`	`NotNull()`	FluentValidation default

Rule-level cascade is not set, so a blank timezone reports both "Timezone cannot be blank." and "Unknown timezone."; the screen shows the first message of a field. Client mirror: `requiredText` (trimmed) and `Validators.maxLength(100)` on `name` → "Application name is required and must be 100 characters or fewer."; `requiredText` on `timezone` → "Timezone is required." The screen trims `name` and `timezone` before sending.

Error cases

Case	Exception / result	Status
Not signed in	JWT bearer challenge	401
PUT without `settings.application.manage`	Policy refusal (logged)	403
Invalid / missing field	`ValidationFilter`	400
Save race	`ConcurrencyConflictException` / `UniqueConstraintException`	409
Stored timezone unknown at run time	Logged; UTC used	–

Frontend design

Member	Behaviour
`ApplicationSettingsComponent.form`	`name` (`requiredText`, max 100), `timezone` (`requiredText`), `defaultPriority` (required, `TASK_PRIORITIES`), `inAppNotificationsEnabled`
`timezoneSuggestions`	UTC, Asia/Kolkata, Asia/Dubai, Europe/London, Europe/Berlin, America/New_York, America/Chicago, America/Los_Angeles, Australia/Sydney (datalist; free text allowed)
`load()`	Loading placeholder; error card "Could not load the application settings." (or server `detail`) with **Retry**
`save()`	Trims; `saving` ("Saving..."); success → `apply`, `savedAt`, toast "Application settings saved"; error → `applyServerErrors` + "Could not save the application settings."
Reminder defaults link	`RouterLink` to Reminder Settings; no reminder fields here (FR-APP-14)

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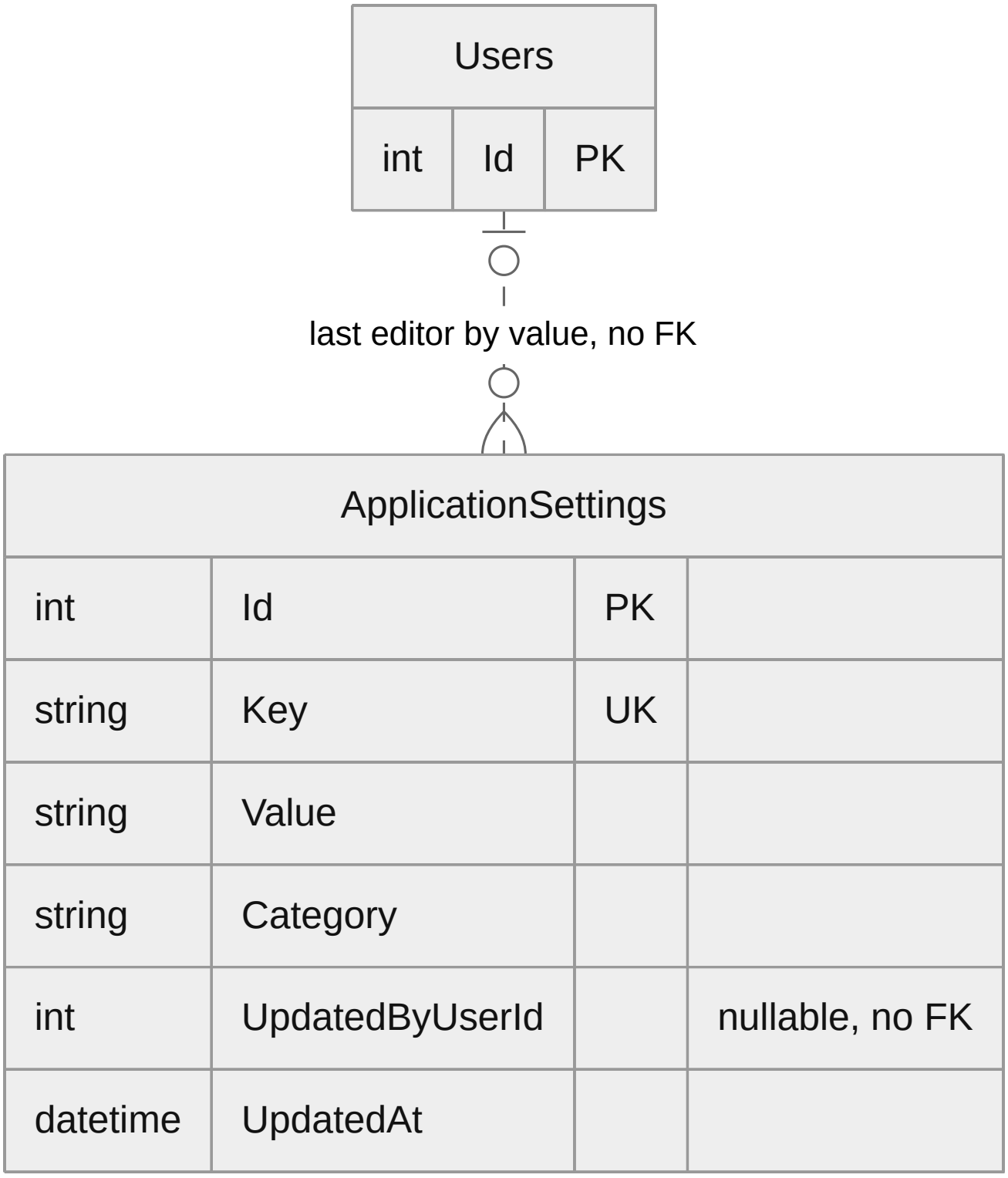
Member	Behaviour
`ApplicationSettingsService.appName`	Signal updated by `get()` and `save()` (`applyName`, fallback "TaskFlow"); the shell binds to it, so the new name shows without reload (FR-APP-05)
Create Task `loadDefaultPriority()`	Pre-selects `defaultPriority` while the field is untouched (FR-APP-11); the server does not apply this default

State machine: *Not applicable for this sub-module.*

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5. Database Design

5.1 ER Diagram



5.2 Tables

`ApplicationSettings` (ApplicationSetting) – shared with Reminder Settings

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Column	Type	Null	Key / index	Default	Description
Id	int (INTEGER)	No	PK	identity	Row id
Key	string(128) (TEXT)	No	Unique `IX_ApplicationSettings_Key`	–	Setting key
Value	string(1024) (TEXT)	No	–	–	Invariant-culture string
Category	string(64) (TEXT)	No	–	–	`Application` or `Notification` for this sub-module
UpdatedByUserId	int? (INTEGER)	Yes	– (no FK)	null	Last editor
UpdatedAt	DateTime (TEXT)	No	–	–	UTC

Rows owned by this sub-module:

Key	Category	Seed default	Read by
`Application.Name`	Application	`TaskFlow`	Shell, `AssistantService` (brand name)
`Application.Timezone`	Application	config `Application:Timezone`, or `UTC` when empty	`AppClock`; shown read-only on Reminder Settings
`Application.DefaultPriority`	Application	`Medium`	Create Task screen
`Notification.InAppEnabled`	Notification	`true`	`NotificationPublisher.InAppEnabledAsync` (missing/unreadable = enabled)

5.3 Indexes, Constraints & Triggers

Object	Definition	Purpose
`IX_ApplicationSettings_Key`	Unique (`Key`)	One row per key
–	No FK, no trigger, no concurrency token	Last writer wins

5.4 Enumerations

Enum	Values	Stored as
`TaskPriority`	`Low`, `Medium`, `High`, `Critical`	Enum name in `ApplicationSettings.Value`; JSON string

5.5 Migrations

Migration	Change
`20261017062735_AddReminderEngine`	Creates `ApplicationSettings` and `IX_ApplicationSettings_Key`

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Migration	Change
`20261017090431_AddInternalMail`	Removes the superseded `Email.%` and `Notification.EmailEnabled` rows

6. Algorithms & Background Jobs

Job / algorithm	Trigger / schedule	Steps	Idempotency
`SettingsInitializer.InitializeAsync`	Every start-up after `MigrateAsync`	Inserts each missing default row; `Application.Timezone` from `ApplicationOptions.Timezone` (or `UTC`)	Existing rows never overwritten
`AppClock.GetTimeZoneAsync`	Each call by a consumer	Read row → resolve id → UTC fallback with error log	Stateless
In-app switch	`NotificationPublisher` before each bell notification	`GetValueAsync(Notification.InAppEnabled)`; returns without writing when false	Stateless

No background job belongs to this sub-module.

7. Security Design (detailed)

Item	Implementation
Read access	`[Authorize]` only: brand name and default priority are needed by every screen; DTO has no secret (FR-APP-18/19). AI keys live in `AiProviderConfigs`, not here
Write access	`[RequirePermission(PermissionKeys.SettingsApplication)]` → `perm:settings.application.manage` (FR-APP-23)
Defaults	`settings.application.manage`: Administrator granted; Manager and Employee not granted on first seed; custom roles configurable
Frontend	`permissionGuard` with `Permissions.SettingsApplication` on `settings/application`; menu item only when held (FR-APP-22)
Timezone input	Validated against the host timezone database before save; never executed
Audit	Denied PUTs logged by `DeniedActionAuditHandler`; saves store `UpdatedByUserId` / `UpdatedAt` only

8. Configuration

Key	Default	Purpose
`Application:Timezone`	`Asia/Kolkata` (code default `UTC`)	Seeds `Application.Timezone`; fallback while the row is missing or blank

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Key	Default	Purpose
`ConnectionStrings:Default`	`<secret>`	SQLite database
`Storage:Root`	`""`	Root folder for the database

9. Testing Design

Test file names from `tests-filelist.json`; contents were not available, coverage inferred from names.

Area	Test class / file	Type	Covers
Service	`TaskFlow.UnitTests/Settings/ApplicationSettings/General/ApplicationSettingsServiceTests.cs`	Unit	Defaults and save
Validator	`TaskFlow.UnitTests/Settings/ApplicationSettings/General/ApplicationSettingsValidatorTests.cs`	Unit	Name and timezone rules
Clock	`TaskFlow.UnitTests/Settings/ApplicationSettings/Clock/AppClockTests.cs`	Unit	Timezone resolution, UTC fallback
Endpoints	`TaskFlow.IntegrationTests/Settings/ApplicationSettings/ApplicationSettingsEndpointsTests.cs`	Integration	GET open to signed-in users, PUT authorization
Seeder	`TaskFlow.UnitTests/Persistence/SettingsInitializerTests.cs`	Unit	Missing rows inserted
Policies	`TaskFlow.IntegrationTests/Common/PermissionAuthorizationTests.cs`	Integration	Policy per key
Support	`TaskFlow.UnitTests/TestSupport/SettingsTestDatabase.cs`, `FakeTimeProvider.cs`	Support	Test DB, clock
Structure	`TaskFlow.UnitTests/Architecture/ArchitectureTests.cs`	Unit	Layer rules

Frontend spec files: 0.

10. Known Limitations & Open Issues

- **In-app switch vs reminders:** `ReminderService` writes deadline/overdue bell notifications through `INotificationRepository` directly, not through `NotificationPublisher`, so they are still created while `Notification.InAppEnabled` is false. FR-APP-13 ("no new bell notification for any event") is therefore not met for reminder notifications.

  > ⚠️ To be confirmed: whether reminder bell notifications should also follow the in-app switch.

- Default task priority is applied only by the Create Task screen, not by `TaskService`.

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- Name length (100) is checked before trimming, the stored value is trimmed; the server does not trim the timezone (the screen does).
- A blank timezone returns two messages ("Timezone cannot be blank." and "Unknown timezone.").
- Accepted timezone ids depend on the host OS timezone database (Windows or IANA ids).
- Stale comments: `ApplicationSettingsController` and `ApplicationSettingsService` (Angular) say "Administrator only"; the code checks `settings.application.manage`.
- No history of earlier values, no reset to defaults, no concurrency token.
- Original BRD Screen 20 "default task reminder intervals" are kept on Reminder Settings (link only, FR-APP-14).

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11. Traceability (FRD → code)

FRD requirement	Endpoint / class	Test
FR-APP-01	`GET /api/v1/settings/application` · `ApplicationSettingsService.GetAsync` · `ApplicationSettingsComponent.load`	`ApplicationSettingsEndpointsTests`, `ApplicationSettingsServiceTests`
FR-APP-02	`ApplicationSettingsComponent.load` (`loadError`, Retry)	– (no frontend specs)
FR-APP-03	`PUT /api/v1/settings/application` · `UpdateAsync` (`Application.Name`)	`ApplicationSettingsServiceTests`
FR-APP-04	Validator `Name` rules · `name.Trim()` · `requiredText`	`ApplicationSettingsValidatorTests`
FR-APP-05	Angular `ApplicationSettingsService.appName` · shell	– (no frontend specs)
FR-APP-06	`timezoneSuggestions` · validator `BeKnownTimezone`	`ApplicationSettingsValidatorTests`
FR-APP-07	Validator `Timezone` rules · `fieldError('timezone')`	`ApplicationSettingsValidatorTests`
FR-APP-08	`AppClock.GetTimeZoneAsync` / `TodayLocalAsync` (read per call)	`AppClockTests`
FR-APP-09	`AppClock` catch → `TimeZoneInfo.Utc` + `LogError`	`AppClockTests`
FR-APP-10	`TaskPriority` · validator `IsInEnum`	`ApplicationSettingsValidatorTests`
FR-APP-11	Create Task `loadDefaultPriority()`	– (no frontend specs)
FR-APP-12	`UpdateAsync` (`Notification.InAppEnabled`)	`ApplicationSettingsServiceTests`
FR-APP-13	`NotificationPublisher.InAppEnabledAsync` (gap for reminders, see §10)	– (Notification tests by name not identified)
FR-APP-14	Reminder defaults `RouterLink` in the page	– (no frontend specs)

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FRD requirement	Endpoint / class	Test
FR-APP-15	`UpdateApplicationSettingsRequestValidator` via `ValidationFilter`	`ApplicationSettingsValidatorTests`, `ApplicationSettingsEndpointsTests`
FR-APP-16	Validator `NotEmpty` / `NotNull` · `ArgumentException` guard	`ApplicationSettingsValidatorTests`
FR-APP-17	`SettingsInitializer` · `ReadString` / `ReadPriority` / `ReadBool`	`SettingsInitializerTests`, `ApplicationSettingsServiceTests`
FR-APP-18	`[Authorize]` on `Get`	`ApplicationSettingsEndpointsTests`
FR-APP-19	`ApplicationSettingsDto` (four fields only)	`ApplicationSettingsEndpointsTests`
FR-APP-20	`Set()` writes `UpdatedByUserId`, `UpdatedAt`	`ApplicationSettingsServiceTests`
FR-APP-21	`ApplicationSettingsComponent.save` (`saving`, toast, `saveError`)	– (no frontend specs)
FR-APP-22	`permissionGuard` (`settings/application`) · `buildNavigation()`	– (no frontend specs)
FR-APP-23	`[RequirePermission(PermissionKeys.SettingsApplication)]` on `Update`	`ApplicationSettingsEndpointsTests`, `PermissionAuthorizationTests`

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